

► CARDIOVASCULAR—PATHOLOGY

Congenital heart diseases

RIGHT-TO-LEFT SHUNTS

Early cyanosis—“blue babies.” Often diagnosed prenatally or become evident immediately after birth. Usually require urgent surgical treatment and/or maintenance of a PDA via prostaglandin therapy.

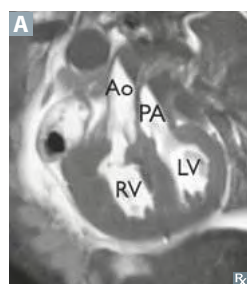
The **5 T**'s:

1. **T**runcus arteriosus (**1** vessel)
2. **T**ransposition (**2** switched vessels)
3. **T**ricuspid atresia (**3** = **Tri**)
4. **T**etralogy of Fallot (**4** = **Tetra**)
5. **TAPVR** (**5** letters in the name)

Persistent truncus arteriosus

Truncus arteriosus fails to divide into pulmonary trunk and aorta due to failure of aorticopulmonary septum formation; most patients have accompanying VSD.

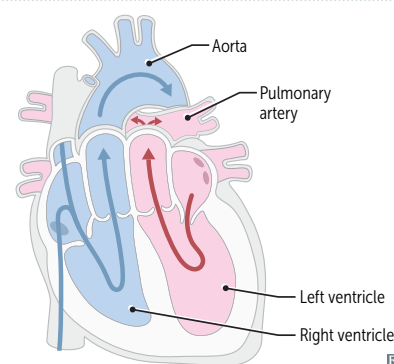
D-transposition of great arteries



Aorta leaves RV (anterior) and pulmonary trunk leaves LV (posterior) → separation of systemic and pulmonary circulations **A**. Not compatible with life unless a shunt is present to allow mixing of blood (eg, VSD, PDA, or patent foramen ovale).

Due to failure of the aorticopulmonary septum to spiral (narrow superior mediastinum causes “egg on a string” appearance on CXR).

Without surgical intervention, most infants die within the first few months of life.

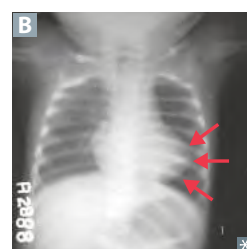


Tricuspid atresia

Absence of tricuspid valve, hypoplastic RV; requires both ASD and VSD/PDA for viability.

ECG shows hypertrophy of RA (tall P-waves) and LV (left axis deviation).

Tetralogy of Fallot



Caused by anterosuperior displacement of the infundibular septum. Most common cause of early childhood cyanosis.

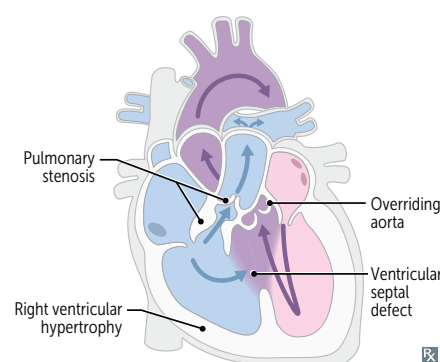
- 1 **P**ulmonary infundibular stenosis (most important determinant for prognosis)
- 2 **R**ight ventricular hypertrophy (RVH)—boot-shaped heart on CXR **B**
- 3 **O**verriding aorta—straddles VSD, receives blood from both LV and RV
- 4 **V**SD

Pulmonary stenosis forces right-to-left flow across VSD → RVH, “tet spells” (often caused by crying, fever, and exercise due to exacerbation of RV outflow obstruction).

PROVe.

Squatting: ↑ SVR, ↓ right-to-left shunt, improves cyanosis.

Associated with 22q11 syndromes.



Total anomalous pulmonary venous return

Pulmonary veins drain into right heart circulation (SVC, coronary sinus, etc); associated with ASD and sometimes PDA to allow for right-to-left shunting to maintain CO.

Ebstein anomaly

Displacement of tricuspid valve leaflets downward into RV, artificially “atrializing” the ventricle. Associated with tricuspid regurgitation, accessory conduction pathways, right-sided HF.

Rare. Can be caused by lithium exposure in utero.

Congenital heart diseases (continued)**LEFT-TO-RIGHT SHUNTS**

Acyanotic at presentation; cyanosis may occur years later. Frequency: VSD > ASD > PDA.

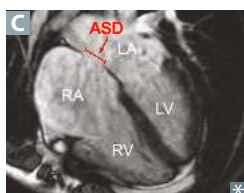
Right-to-left shunts: early cyanosis.

Left-to-right shunts: “later” cyanosis.

Ventricular septal defect

Asymptomatic at birth, may manifest weeks later or remain asymptomatic throughout life. Most smaller defects self-resolve; larger defects, if left surgically untreated, cause ↑ pulmonary blood flow and LV overload (Eisenmenger syndrome), which may progress to HF.

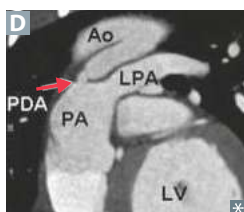
O₂ saturation ↑ in RV and pulmonary artery.

Atrial septal defect

Defect in interatrial septum **C**; systolic ejection murmur with wide, fixed split S2. Ostium secundum defects most common and usually an isolated finding; ostium primum defects rarer and usually occur with other cardiac anomalies. Symptoms range from none to HF. Distinct from patent foramen ovale, which is due to failed fusion.

O₂ saturation ↑ in RA, RV, and pulmonary artery. May lead to paradoxical emboli (systemic venous emboli use ASD to bypass lungs and become systemic arterial emboli) in the setting of temporary shunt reversal (eg, when lifting weights or in Eisenmenger syndrome).

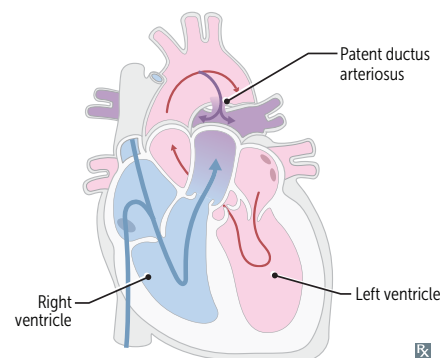
Associated with Down syndrome.

Patent ductus arteriosus

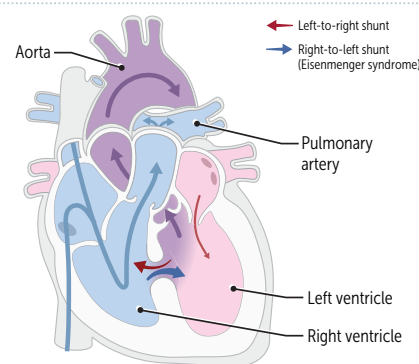
In fetal period, shunt is right to left (normal). In neonatal period, ↓ pulmonary vascular resistance → shunt becomes left to right → progressive RVH and/or LVH and HF.

Associated with a continuous, “machinelike” murmur. Patency is maintained by PGE synthesis and low O₂ tension. Uncorrected PDA **D** can eventually result in late cyanosis in the lower extremities (differential cyanosis).

PDA is normal in utero and normally closes soon after birth.

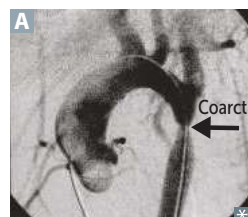
**Eisenmenger syndrome**

Uncorrected left-to-right shunt (VSD, ASD, PDA) → ↑ pulmonary blood flow → pathologic remodeling of vasculature → pulmonary arterial hypertension. RVH occurs to compensate → shunt becomes right to left when RV > LV pressure (see illustration). Causes late cyanosis, clubbing, and polycythemia. Age of onset varies depending on size and severity of initial left-to-right shunt.

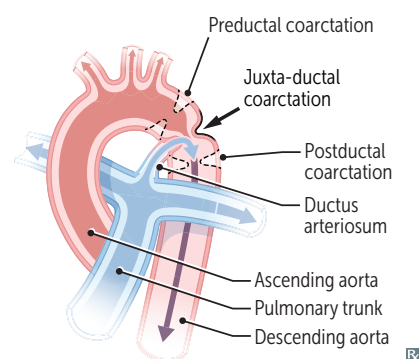


Eisenmenger syndrome from chronic VSD

Coarctation of the aorta



Aortic narrowing **A** near insertion of ductus arteriosus (“juxtaductal”). Associated with bicuspid aortic valve, other heart defects, and Turner syndrome. Hypertension in upper extremities. Cyanosis, claudication, coolness, and weak, delayed pulses (brachiofemoral delay) in lower extremities. With age, intercostal arteries enlarge due to collateral circulation → rib-notching on x-ray. Complications include HF, ↑ risk of cerebral hemorrhage (berry aneurysms), aortic rupture, and infective endocarditis.



Persistent pulmonary hypertension of the newborn

Persistence of ↑ pulmonary vascular resistance after birth. Associated with abnormal development and postpartum adaptation of pulmonary vasculature. Risk factors include aspiration of meconium-stained amniotic fluid and neonatal pneumonia. Leads to right-to-left shunt through foramen ovale and ductus arteriosus. Preductal O₂ saturation is often > postductal. Newborn presents with signs of respiratory distress (eg, tachypnea) and (often differential) cyanosis. Equal pulses (no delay).

Congenital cardiac defect associations

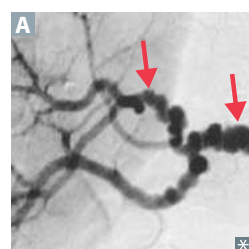
ASSOCIATION	DEFECT
Prenatal alcohol exposure (fetal alcohol syndrome)	VSD, PDA, ASD, tetralogy of Fallot
Congenital rubella	PDA, pulmonary artery stenosis, septal defects
Down syndrome	AVSD, VSD, ASD
Infant of patient with diabetes during pregnancy	TGA, truncus arteriosus, tricuspid atresia, VSD
Marfan syndrome	MVP, thoracic aortic aneurysm/dissection, AR
Prenatal lithium exposure	Ebstein anomaly
Turner syndrome	Bicuspid aortic valve, aortic coarctation/dissection
Williams syndrome	Supravalvular aortic stenosis
22q11 syndromes	Truncus arteriosus, tetralogy of Fallot

Hypertension

RISK FACTORS

Persistent systolic BP ≥ 130 mm Hg and/or diastolic BP ≥ 80 mm Hg.

FEATURES



↑ age, obesity, diabetes, physical inactivity, high-sodium diet, excess alcohol intake, tobacco smoking, family history; incidence greatest in Black > White > Asian populations.

90% of hypertension is 1° (essential) and related to ↑ CO or ↑ TPR. Remaining 10% mostly 2° to renal/renovascular diseases such as fibromuscular dysplasia (characteristic “string of beads” appearance of renal artery **A**, usually seen in adult females) and atherosclerotic renal artery stenosis, 1° hyperaldosteronism, or obstructive sleep apnea.

Hypertensive urgency—severe (≥ 180/≥ 120 mm Hg) hypertension without acute end-organ damage.

Hypertensive emergency—severe hypertension with evidence of acute end-organ damage (eg, encephalopathy, stroke, retinal hemorrhages and exudates, papilledema, MI, HF, aortic dissection, kidney injury, microangiopathic hemolytic anemia, eclampsia). Arterioles may show fibrinoid necrosis.

PREDISPOSES TO

CAD, concentric LVH (mediated by angiotensin II and endothelin), HF, atrial fibrillation; aortic dissection/aneurysm; stroke; CKD (hypertensive nephropathy); retinopathy.

Hyperlipidemia signs

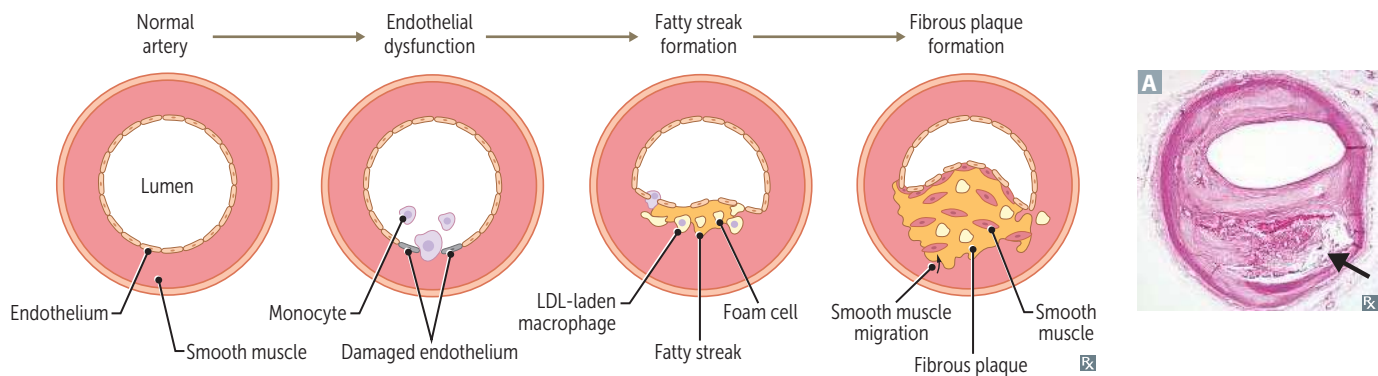
Xanthomas	Plaques or nodules composed of lipid-laden histiocytes in skin A , especially the eyelids (xanthelasma B).
Tendinous xanthoma	Lipid deposit in tendon C , especially Achilles tendon and finger extensors. Associated with familial hypercholesterolemia.
Corneal arcus	Lipid deposit in cornea. Common in older adults (arcus senilis D), but appears earlier in life with hypercholesterolemia.



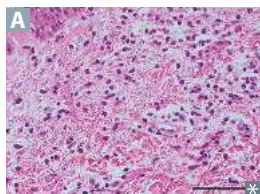
Atherosclerosis

Very common form of arteriosclerosis (hardening of arteries). Disease of elastic arteries and large- and medium-sized muscular arteries; caused by buildup of cholesterol plaques in tunica intima.

LOCATION	Abdominal aorta > coronary artery > popliteal artery > carotid artery > circle of Willis. A copy cat named Willis.
RISK FACTORS	Modifiable: hypertension, tobacco smoking, dyslipidemia (↑ LDL, ↓ HDL), diabetes. Non-modifiable: age, male sex, postmenopausal status, family history.
SYMPTOMS	Angina, claudication, but can be asymptomatic.
PROGRESSION	Inflammation important in pathogenesis: endothelial cell dysfunction → macrophage and LDL accumulation → foam cell formation → fatty streaks → smooth muscle cell migration (involves PDGF and FGF), proliferation, and extracellular matrix deposition → fibrous plaque → complex atheromas A → calcification (calcium content correlates with risk of complications).
COMPLICATIONS	Ischemia, infarction, aneurysm formation, peripheral vascular disease, thrombosis, embolism, renovascular hypertension, coarctation of the aorta, subclavian steal syndrome.



Cholesterol emboli syndrome

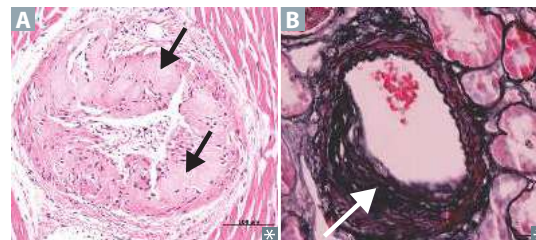


Microembolization of cholesterol displaced from atherosclerotic plaques **A** in large arteries (usually the aorta). Results in end-organ damage due to small artery emboli and an inflammatory response (eg, livedo reticularis, digital ischemia [blue toe syndrome], acute renal failure, cerebrovascular accident, gut ischemia). Pulses remain palpable because larger arteries are unaffected. May follow invasive vascular procedures (angiography, angioplasty, endovascular grafting).

Arteriolosclerosis

Common form of arteriosclerosis. Affects small arteries and arterioles. Two types:

- **Hyaline**—vessel wall thickening 2° to plasma protein leak into subendothelium in hypertension or diabetes mellitus **A**.
- **Hyperplastic**—“onion skinning” **B** in severe hypertension with proliferation of smooth muscle cells.



Aortic aneurysm

Localized pathologic dilation of the aorta. May cause abdominal and/or back pain, which is a sign of leaking, dissection, or imminent rupture.

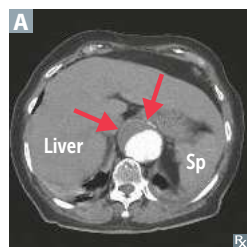
Thoracic aortic aneurysm

Associated with cystic medial degeneration. Risk factors include hypertension, bicuspid aortic valve, connective tissue disease (eg, Marfan syndrome). Also associated with 3° syphilis (obliterative endarteritis of the vasa vasorum). Aortic root dilatation may lead to aortic valve regurgitation.



- Ascending thoracic aorta
- Aortic arch
- Descending thoracic aorta
- Abdominal aorta

Abdominal aortic aneurysm

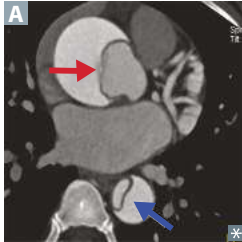


Associated with transmural (all 3 layers) inflammation and extracellular matrix degradation. Risk factors include tobacco smoking (strongest risk factor), ↑ age, male sex, family history. May present as palpable pulsatile abdominal mass (arrows in **A** point to outer dilated aortic wall). Rupture may present as triad of pulsatile abdominal mass, acute abdominal/back pain, and resistant hypotension. Most often infrarenal (distribution of vasa vasorum is reduced).

Traumatic aortic rupture

Due to trauma and/or deceleration injury (MVA or significant fall), most commonly at aortic isthmus (proximal descending aorta just distal to origin of left subclavian artery). X-ray may reveal widened mediastinum.

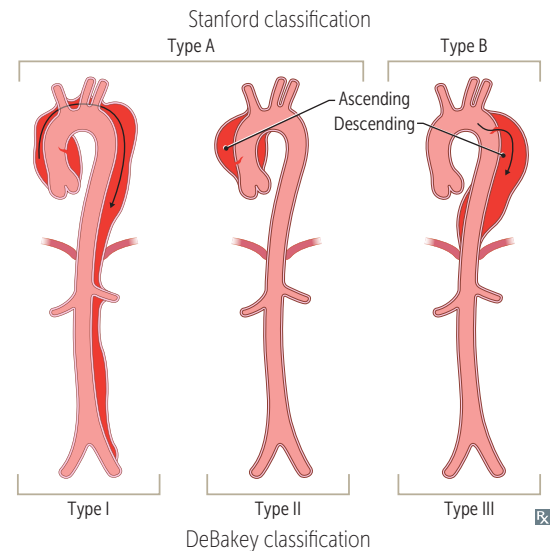
Aortic dissection



Longitudinal intimal tear forming a false lumen. Associated with hypertension (strongest risk factor), bicuspid aortic valve, inherited connective tissue disorders (eg, Marfan syndrome). Can present with tearing, sudden-onset chest pain radiating to the back +/- markedly unequal BP in arms. CXR can show mediastinal widening. Can result in organ ischemia, embolic stroke, aortic rupture, death.

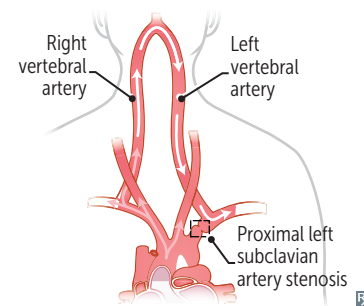
Stanford type **A** (proximal): involves **A**scending aorta (red arrow in **A**). May extend to aortic arch or descending aorta (blue arrow in **A**). May result in acute aortic regurgitation or cardiac tamponade. Treatment: surgery.

Stanford type **B** (distal): involves only descending aorta (**B**elow left subclavian artery). Treatment: β -blockers, then vasodilators.

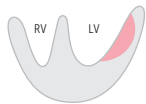
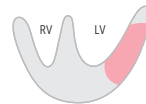


Subclavian steal syndrome

Stenosis of subclavian artery proximal to origin of vertebral artery → hypoperfusion distal to stenosis → reversed blood flow in ipsilateral vertebral artery → reduced cerebral perfusion on exertion of affected arm. Causes arm ischemia, pain, paresthesia, vertebrobasilar insufficiency (dizziness, vertigo), > 15 mm Hg difference in systolic BP between arms. Associated with atherosclerosis (most common cause), Takayasu arteritis, heart surgery.



Coronary artery disease

Angina	Chest pain due to ischemic myocardium 2° to coronary artery narrowing or spasm; no necrosis. ▪ Stable—usually 2° to atherosclerosis ($\geq 70\%$ occlusion); exertional chest pain in classic distribution resolving with rest or nitroglycerin. ▪ Unstable—thrombosis with incomplete coronary artery occlusion; ↑ in frequency or intensity of exertional chest pain or any chest pain at rest. No cardiac biomarker elevation (vs non-ST-segment elevation MI [NSTEMI]). ▪ Vasospastic (formerly Prinzmetal or variant)—occurs at rest 2° to coronary artery spasm; transient ischemic ST changes on ECG. Tobacco smoking is a major risk factor. Triggers include cocaine, amphetamines, alcohol, triptans. Treat with Ca^{2+} channel blockers, nitrates, and smoking cessation (if applicable).			
Myocardial infarction	Most often due to an acute coronary syndrome (ACS): rupture of coronary artery atherosclerotic plaque → acute thrombosis. MI can also occur with prolonged supply-demand mismatch (eg, stable angina → prolonged tachycardia and hypotension from pneumonia → elevated troponin but no acute plaque rupture).			
	Stable angina	Unstable angina	NSTEMI	STEMI
PAIN	On exertion	Mild exertion or at rest	At rest	At rest
TROPONIN LEVEL	No elevation	No elevation	Elevated	Elevated
INFARCTION	None	None	 Subendocardial 	 Transmural 
ECG CHANGES	Possible ST depression and/or T-wave inversion	Possible ST depression and/or T-wave inversion	ST depression and/or T-wave inversion	ST elevation, pathologic Q waves

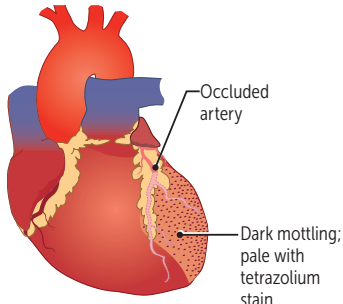
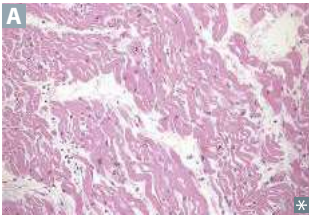
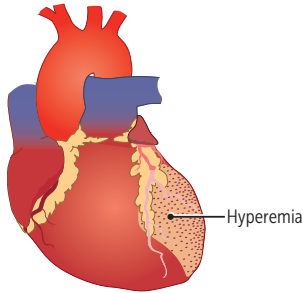
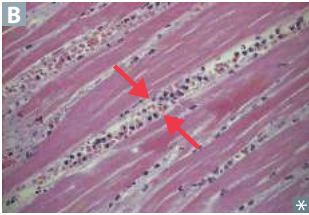
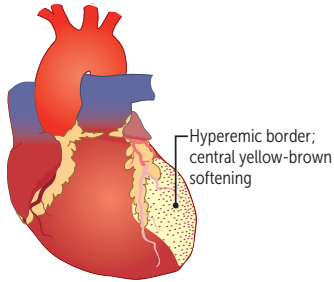
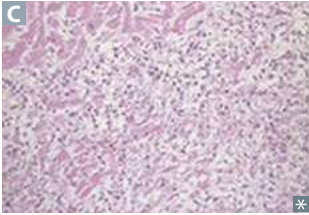
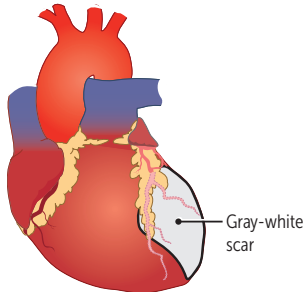
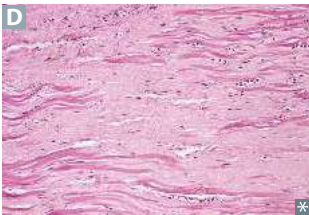
Ischemic heart disease manifestations

Coronary steal syndrome	Distal to coronary stenosis, vessels are maximally dilated at baseline to compensate for reduced blood flow. Administration of vasodilators (eg, dipyridamole, adenosine, regadenoson) dilates normal vessels → ↓ hydrostatic pressure in normal coronary arteries → blood is shunted toward well-perfused areas → ↓ flow to myocardium perfused by stenosed vessels (“steal”) → ischemia of myocardium downstream to pathologically dilated vessels. Vasodilator stress tests rely on differential in flow to detect potential ischemia. Rarely, they can cause coronary steal and true ischemia. Vasodilation of healthy vessels steals blood from stenosed vessels.
Sudden cardiac death	Unexpected death due to cardiac causes within 1 hour of symptom onset or within 24 hours with no cardiovascular symptoms, most commonly due to lethal ventricular arrhythmia (eg, ventricular fibrillation) impairing blood flow to the brain. Associated with CAD (up to 70% of cases), cardiomyopathy (hypertrophic, dilated), myocarditis, coronary artery anomalies, and hereditary channelopathies (eg, long QT syndrome, Brugada syndrome). Prevent with implantable cardioverter-defibrillator.
Chronic ischemic heart disease	Progressive exertional symptoms and/or development of HF due to chronic ischemic myocardial damage. Myocardial hibernation —LV systolic dysfunction in the setting of chronic ischemia. Potentially reversible with myocardial reperfusion. Seen in stable angina, acute MI, or HF. Contrast with myocardial stunning —transient, reversible LV systolic dysfunction after brief, acute ischemia.

Evolution of myocardial infarction

Commonly occluded coronary arteries: LAD > RCA > circumflex.

Symptoms: diaphoresis, nausea, vomiting, severe retrosternal pain, pain in left arm and/or jaw, shortness of breath, fatigue.

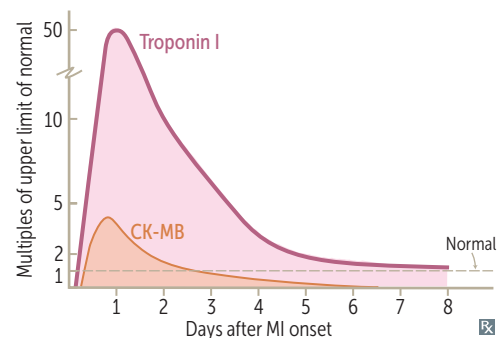
TIME	GROSS	LIGHT MICROSCOPE	COMPLICATIONS
0–24 hours	 Occluded artery Dark mottling; pale with tetrazolium stain	Wavy fibers (0–4 hr), early coagulative necrosis (4–24 hr) A → cell content released into blood; edema, hemorrhage Reperfusion injury → free radicals and ↑ Ca^{2+} influx → hypercontraction of myofibrils (dark eosinophilic stripes) 	Ventricular arrhythmia, HF, cardiogenic shock
1–3 days	 Hyperemia	Extensive coagulative necrosis Tissue surrounding infarct shows acute inflammation with neutrophils B 	Postinfarction fibrinous pericarditis
3–14 days	 Hyperemic border; central yellow-brown softening	Macrophages, then granulation tissue at margins C 	Free wall rupture → tamponade; papillary muscle rupture → mitral regurgitation; interventricular septal rupture due to macrophage-mediated structural degradation → left-to-right shunt LV pseudoaneurysm (risk of rupture)
2 weeks to several months	 Gray-white scar	Contracted scar complete D 	Postcardiac injury syndrome, HF, arrhythmias, true ventricular aneurysm (risk of mural thrombus)

Diagnosis of myocardial infarction

In the first 6 hours, ECG is the gold standard. Cardiac troponin **I** rises after 4 hours (peaks at 24 hr) and is $\uparrow$ for 7–10 days; more specific than other protein markers.

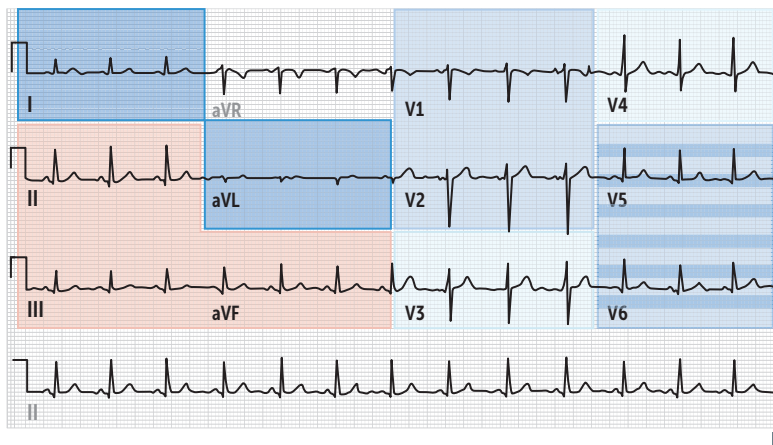
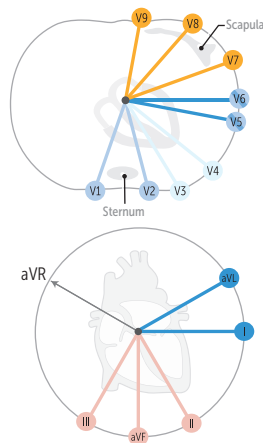
CK-MB increases after 6–12 hours (peaks at 16–24 hr) and is predominantly found in myocardium but can also be released from skeletal muscle. Useful in diagnosing reinfarction following acute MI because levels return to normal after 48 hours.

ECG changes can include ST elevation (STEMI, transmural infarct), ST depression (NSTEMI, subendocardial infarct), hyperacute (peaked) T waves, T-wave inversion, and pathologic Q waves or poor R wave progression (evolving or old transmural infarct).



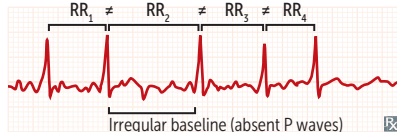
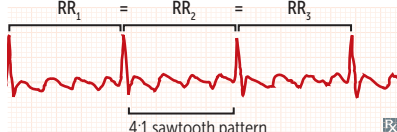
ECG localization of STEMI

INFARCT LOCATION	LEADS WITH ST-SEGMENT ELEVATIONS OR Q WAVES
Anteroseptal (LAD)	V ₁ –V ₂
Anterapical (distal LAD)	V ₃ –V ₄
Anterolateral (LAD or LCX)	V ₅ –V ₆
Lateral (LCX)	I, aVL
InFerior (RCA)	II, III, aVF
Posterior (PDA)	V ₇ –V ₉ , ST depression in V ₁ –V ₃ with tall R waves



Narrow complex tachycardias

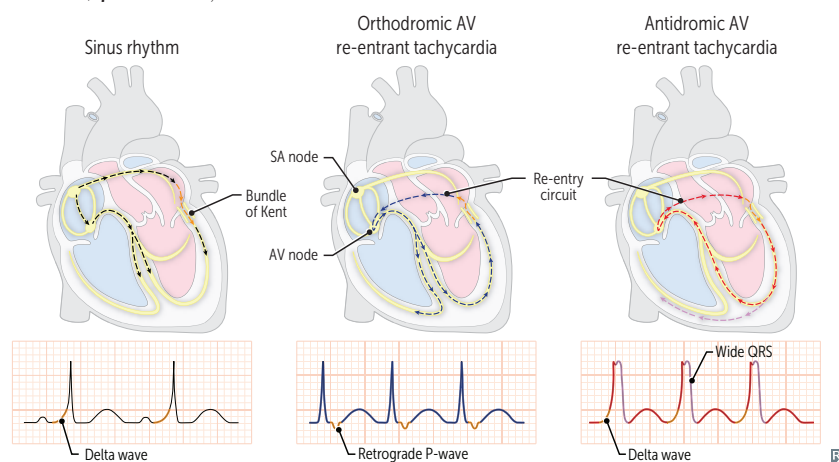
Narrow QRS complex < 120 msec, rapid ventricular activation via normal ventricular conduction system, tachycardia originates within or above AV node (supraventricular arrhythmia).

ARRHYTHMIA	DESCRIPTION	ECG FINDINGS
Atrial fibrillation	Irregularly irregular rate and rhythm with no discrete P waves. Arrhythmogenic activity usually originates from automatic foci near pulmonary vein ostia in left atrium. Common risk factors include hypertension, CAD, advanced age, atrial dilation. May predispose to thromboembolic events due to LA blood stasis, particularly stroke. LA appendage is the most common site of thrombus formation in atrial fibrillation. Management of Atrial fibrillation involves rate and rhythm control using Beta-Blockers, Calcium Channel blockers, and Digoxin (ABCD) and cardioversion.	 Irregular baseline (absent P waves)
Multifocal atrial tachycardia	Irregularly irregular rate and rhythm with at least 3 distinct P wave morphologies, due to multiple ectopic foci in atria. Associated with underlying conditions such as COPD, pneumonia, HF.	
Atrial flutter	Rapid succession of identical, consecutive atrial depolarization waves → “sawtooth” appearance of P waves. Arrhythmogenic activity usually originates from reentry circuit around tricuspid annulus. Treat like atrial fibrillation +/- catheter ablation of region between tricuspid annulus and IVC.	 4:1 sawtooth pattern
Paroxysmal supraventricular tachycardia	Most often due to a reentrant tract between atrium and ventricle, most commonly in AV node. Commonly presents with sudden-onset palpitations, lightheadedness, diaphoresis. Terminate by slowing AV nodal conduction (vagal maneuvers, adenosine). Definitively treat with catheter ablation.	

Wolff-Parkinson-White syndrome

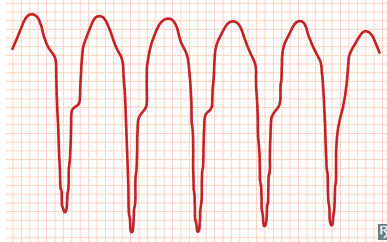
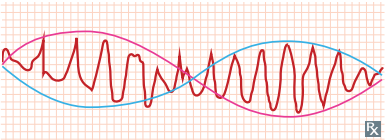
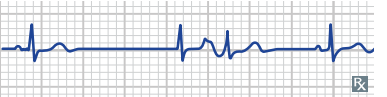
Most common type of ventricular preexcitation syndrome. Abnormal fast accessory conduction pathway from atria to ventricle (bundle of Kent) bypasses rate-slowing AV node → ventricles partially depolarize earlier → characteristic delta wave with widened QRS complex and shortened PR interval. May result in reentry circuit → supraventricular tachycardia.

Treatment: procainamide, ibutilide. Avoid AV nodal-blocking drugs (eg, adenosine, calcium channel blockers, β -blockers).

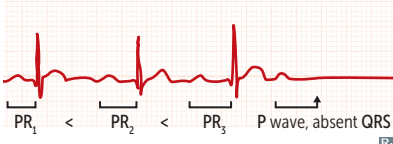
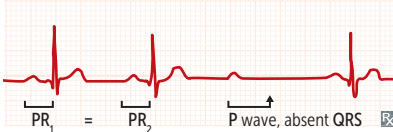
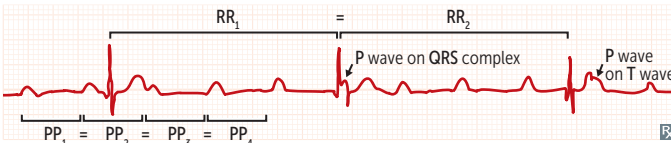
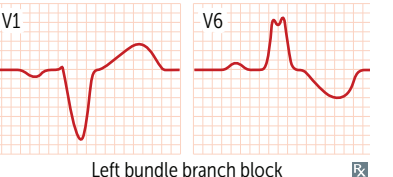


Wide complex tachycardias

Wide QRS complex ≥ 120 msec, slow ventricular activation outside normal ventricular conduction system, tachycardia originates below AV node (ventricular arrhythmia).

ARRHYTHMIA	DESCRIPTION	ECG FINDINGS
Ventricular tachycardia	Typically regular rhythm, rate > 100 . Most commonly due to structural heart disease (eg, cardiomyopathy, scarring after myocardial infarction). High risk of sudden cardiac death.	
Torsades de pointes	Polymorphic ventricular tachycardia. Shifting sinusoidal waveforms. May progress to ventricular fibrillation. Long QT interval (eg, sinus bradycardia, congenital long QT syndromes) predisposes to torsades de pointes. Caused by drugs, $\downarrow K^+$, $\downarrow Mg^{2+}$, $\downarrow Ca^{2+}$. Torsades de pointes = twisting of the points Treatment: defibrillation for unstable patients, magnesium sulfate for stable patients. Drug-induced long QT (ABCDEF+NO): - anti-Arrhythmics (Ia and III), Arsenic - anti-Biotics (macrolides, fluoroquinolones) - anti-Cychotics (haloperidol), Chloroquine - anti-Depressants (TCAs), Diuretics (thiazides) - anti-Emetics (ondansetron) - anti-Fungals (Fluconazole) Navir (protease inhibitors) Opioids (methadone)	
Ventricular fibrillation	Disorganized rhythm with no identifiable waves. Treatment: fatal without immediate CPR and defibrillation.	 No discernible rhythm
Hereditary channelopathies	Inherited mutations of cardiac ion channels $\rightarrow$ abnormal myocardial action potential $\rightarrow$ $\uparrow$ risk of ventricular tachyarrhythmias and sudden cardiac death (SCD).	
Brugada syndrome	Autosomal dominant; most commonly due to loss of function mutation of Na^+ channels. $\uparrow$ prevalence in Asian males. ECG pattern of pseudo-right bundle branch block and ST-segment elevations in leads V_1 – V_2 . Prevent SCD with ICD.	
Congenital long QT syndrome	Most commonly due to loss of function mutation of K^+ channels (affects repolarization). Includes: Romano-Ward syndrome—autosomal dominant, pure cardiac phenotype (no deafness). Jervell and Lange-Nielsen syndrome—autosomal recessive, sensorineural deafness.	
Sick sinus syndrome	Age-related degeneration of SA node. ECG can show bradycardia, sinus pauses (delayed P waves), sinus arrests (dropped P waves), junctional escape beats.	

Conduction blocks

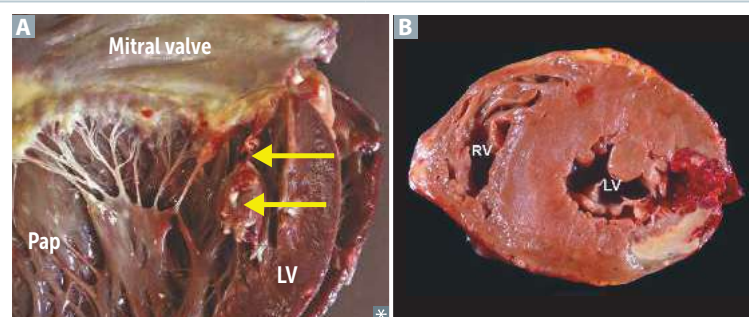
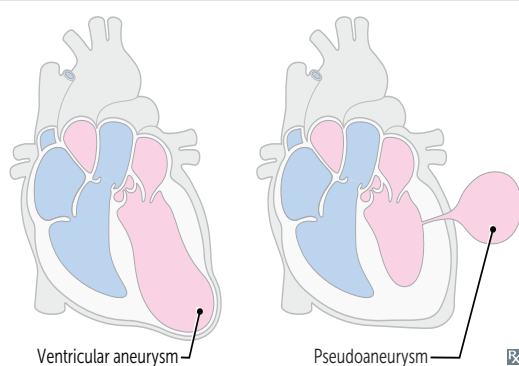
ARRHYTHMIA	DESCRIPTION	ECG FINDINGS
First-degree AV block	Prolonged PR interval (>200 msec). Treatment: none required (benign and asymptomatic).	 PR ₁ = PR ₂ = PR ₃ = PR ₄
Second-degree AV block Mobitz type I (Wenckebach)	Progressive lengthening of PR interval until a beat is “dropped” (P wave not followed by QRS complex). Variable RR interval with a pattern (regularly irregular). Treatment: none required (usually asymptomatic)	 PR ₁ < PR ₂ < PR ₃ P wave, absent QRS
Second-degree AV block Mobitz type II	Dropped beats that are not preceded by a change in PR interval. May progress to 3rd-degree block, as it usually indicates a structural abnormality such as ischemia or fibrosis. Treatment: usually a pacemaker.	 PR ₁ = PR ₂ P wave, absent QRS
Third-degree (complete) AV block	P waves and QRS complexes rhythmically dissociated. Atria and ventricles beat independently of each other. Atrial rate > ventricular rate. May be caused by Lyme disease. Treatment: pacemaker.	 RR ₁ = RR ₂ PP ₁ = PP ₂ = PP ₃ = PP ₄ P wave on QRS complex P wave on T wave
Bundle branch block	Interruption of conduction of normal left or right bundle branches. Affected ventricle depolarizes via slower myocyte-to-myocyte conduction from the unaffected ventricle, which depolarizes via the faster His-Purkinje system. Commonly due to degenerative changes (eg, cardiomyopathy, infiltrative disease).	 V1 V6 Left bundle branch block

Premature beats

ARRHYTHMIA	DESCRIPTION	ECG FINDINGS
Premature atrial contraction	Extra beats arising from ectopic foci in atria instead of the SA node. Often 2° to ↑ adrenergic drive (eg, caffeine consumption). Benign, but may increase risk for atrial fibrillation and flutter. Narrow QRS complex with preceding P wave on ECG.	
Premature ventricular contraction	Ectopic beats arising from ventricle instead of the SA node. Shortened diastolic filling time → ↓ SV compared to a normal beat. Prognosis is largely influenced by underlying heart disease. Wide QRS complex with no preceding P wave on ECG.	

Myocardial infarction complications

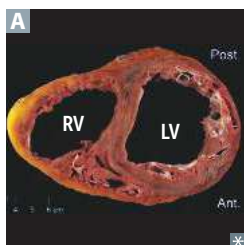
COMPLICATION	TIMEFRAME	FINDINGS	NOTES
Cardiac arrhythmia	First few days to several months	Can be supraventricular arrhythmias, ventricular arrhythmias, or conduction blocks.	Due to myocardial death and scarring. Important cause of death before reaching the hospital and within the first 48 hours post-MI.
Peri-infarction pericarditis	1–3 days	Pleuritic chest pain, pericardial friction rub, ECG changes, and/or small pericardial effusion.	Usually self-limited.
Papillary muscle rupture	2–7 days	Can result in acute mitral regurgitation → cardiogenic shock, severe pulmonary edema.	Posteromedial >> anterolateral papillary muscle rupture A , as the posteromedial has single artery blood supply (posterior descending artery) whereas anterolateral has dual (LAD, LCX).
Interventricular septal rupture	3–5 days	Symptoms can range from mild to severe with cardiogenic shock and pulmonary edema.	Macrophage-mediated degradation → VSD → ↑ O ₂ saturation and ↑ pressure in RV.
Ventricular pseudoaneurysm	3–14 days	May be asymptomatic. Symptoms may include chest pain, murmur, arrhythmia, syncope, HF, embolus from mural thrombus. Rupture → cardiac tamponade.	Free wall rupture contained by adherent pericardium or scar tissue—does not contain endocardium or myocardium. More likely to rupture than true aneurysm.
Ventricular free wall rupture	5–14 days	Free wall rupture B → cardiac tamponade or internal hemorrhage, often fatal.	LV hypertrophy and previous MI protect against free wall rupture.
True ventricular aneurysm	2 weeks to several months	Symptoms similar to pseudoaneurysm.	Outward bulge with contraction (“dyskinesia”). Associated with fibrosis.
Postcardiac injury syndrome	Weeks to several months	Pericarditis due to autoimmune reaction.	Also called Dressler syndrome. Cardiac antigens released after injury → deposition of immune complexes in pericardium → inflammation.



Acute coronary syndrome treatments

Unstable angina/NSTEMI—Anticoagulation (eg, heparin), antiplatelet therapy (eg, aspirin) + ADP receptor inhibitors (eg, clopidogrel), β -blockers, ACE inhibitors, statins. Symptom control with nitroglycerin +/- morphine.

STEMI—In addition to above, reperfusion therapy most important (percutaneous coronary intervention preferred over fibrinolysis). If RV affected (eg, RCA occlusion), support venous return/preload to maintain cardiac output (eg, IV fluids, avoiding nitroglycerin).

Cardiomyopathies**Dilated cardiomyopathy**

Most common cardiomyopathy (90% of cases).

Often idiopathic or familial (eg, due to mutation of *TTN* gene encoding the sarcomeric protein titin).

Other etiologies include drugs (eg, alcohol, cocaine, doxorubicin), infection (eg, coxsackie B virus, Chagas disease), ischemia (eg, CAD), systemic conditions (eg, hemochromatosis, sarcoidosis, thyrotoxicosis, wet beriberi), peripartum cardiomyopathy.

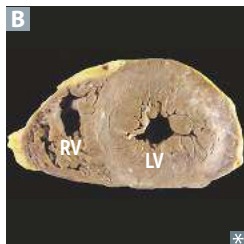
Findings: HF, S3, systolic regurgitant murmur, dilated heart on echocardiogram, balloon appearance of heart on CXR.

Treatment: Na^+ restriction, ACE inhibitors/ARBs, β -blockers, sacubitril, diuretics, mineralocorticoid receptor blockers (eg, spironolactone), ICD, heart transplant.

Systolic dysfunction ensues.

Displays **eccentric hypertrophy** **A** (sarcomeres added in series). Compare to athlete's heart, where LV and RV enlargement facilitates $\uparrow$ SV and $\uparrow$ CO.

Stress cardiomyopathy (also called takotsubo cardiomyopathy, broken heart syndrome)—ventricular apical ballooning likely due to $\uparrow$ sympathetic stimulation (eg, stressful situations).

Hypertrophic cardiomyopathy

60–70% of cases are familial, autosomal dominant (most commonly due to mutations in genes encoding sarcomeric proteins, such as myosin binding protein C and β -myosin heavy chain). Causes syncope during exercise and may lead to sudden death (eg, in young athletes) due to ventricular arrhythmia.

Findings: S4, systolic murmur. May see mitral regurgitation due to impaired mitral valve closure.

Treatment: use of β -blockers or nondihydropyridine Ca^{2+} channel blockers (eg, verapamil) and, in some cases, cessation of high-intensity athletics. ICD if high risk. Avoid drugs that decrease preload (eg, diuretics, vasodilators).

Diastolic dysfunction ensues.

Displays ventricular **concentric hypertrophy** (sarcomeres added in parallel) **B**, often septal predominance. Myofibrillar disarray and fibrosis.

Classified as **hypertrophic obstructive cardiomyopathy** when LV outflow tract is obstructed. Asymmetric septal hypertrophy and systolic anterior motion of mitral valve $\rightarrow$ outflow obstruction $\rightarrow$ dyspnea, possible syncope.

Restrictive/infiltrative cardiomyopathy

Postradiation fibrosis, **L**öffler endocarditis, **E**ndocardial fibroelastosis (thick fibroelastic tissue in endocardium of young children), **A**myloidosis, **S**arcoidosis, **H**emochromatosis (**PLEASE** Help!).

Diastolic dysfunction ensues. Can have low-voltage ECG despite thick myocardium (especially in amyloidosis).

Löffler endocarditis—associated with hypereosinophilic syndrome; histology shows eosinophilic infiltrates in myocardium.

Heart failure

Clinical syndrome of cardiac pump dysfunction → congestion and low perfusion. Symptoms include dyspnea, orthopnea, fatigue; signs include S3 heart sound, rales, jugular venous distention (JVD), pitting edema **A**.

Systolic dysfunction—heart failure with reduced ejection fraction (HFrEF), ↑ EDV; ↓ contractility often 2° to ischemia/MI or dilated cardiomyopathy.

Diastolic dysfunction—heart failure with preserved ejection fraction (HFpEF); ↓ compliance (↑ EDP) often 2° to myocardial hypertrophy.

Right HF most often results from left HF. Cor pulmonale refers to isolated right HF due to pulmonary cause.

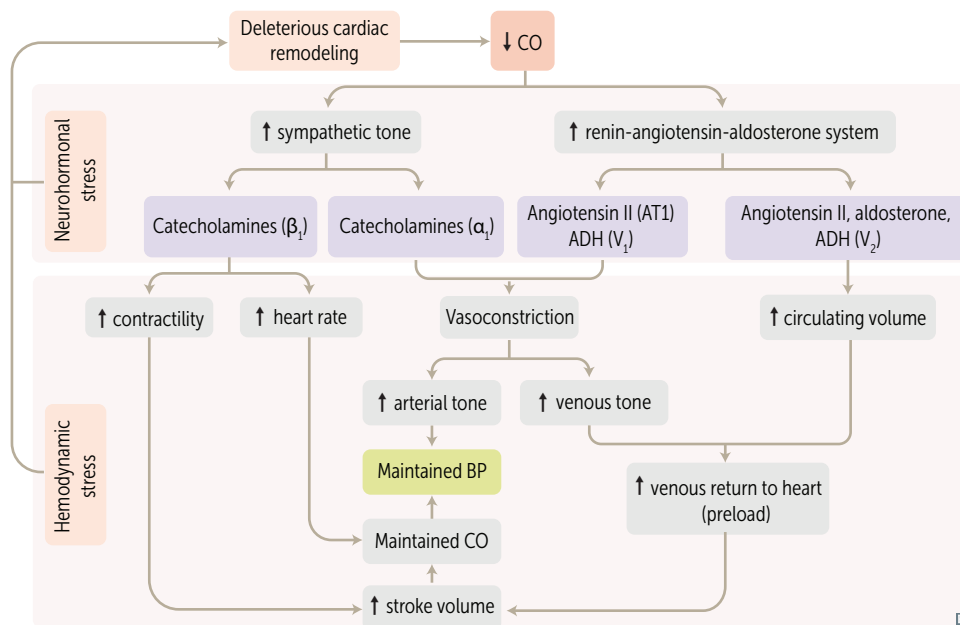
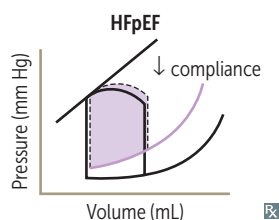
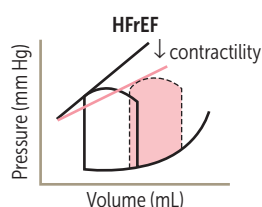
ACE inhibitors, ARBs, angiotensin receptor–neprilysin inhibitors, β-blockers (except in acute decompensated HF), and aldosterone receptor antagonists ↓ mortality in HFrEF. Loop and thiazide diuretics are used mainly for symptomatic relief. Hydralazine with nitrate therapy and SGLT2 inhibitors improve both symptoms and mortality in select patients.

Left heart failure

Orthopnea	Shortness of breath when supine: ↑ venous return from redistribution of blood (immediate gravity effect) exacerbates pulmonary vascular congestion.
Paroxysmal nocturnal dyspnea	Breathless awakening from sleep: ↑ venous return from redistribution of blood, reabsorption of peripheral edema, etc.
Pulmonary edema	↑ pulmonary venous pressure → pulmonary venous distention and transudation of fluid. Presence of hemosiderin-laden macrophages (“HF” cells) in lungs.

Right heart failure

Congestive hepatomegaly	↑ central venous pressure → ↑ resistance to portal flow. Rarely, leads to “cardiac cirrhosis.” Associated with nutmeg liver (mottled appearance) on gross exam.
Jugular venous distention	↑ venous pressure.
Peripheral edema	↑ venous pressure → fluid transudation.



High-output heart failure

Uncommon form of HF characterized by ↑ CO. High-output state is due to ↓ SVR from either vasodilation or arteriovenous shunting. Causes include severe obesity, advanced cirrhosis, severe anemia, hyperthyroidism, wet beriberi, Paget disease of bone. Presents with symptoms and signs of pulmonary and/or systemic venous congestion.

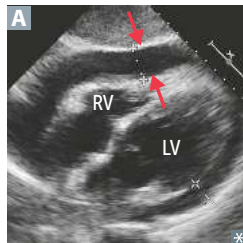
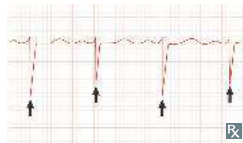
Shock

Inadequate organ perfusion and delivery of nutrients necessary for normal tissue and cellular function. Initially may be reversible but life threatening if not treated promptly.

TYPE	CAUSED BY	MECHANISM	SKIN	CVP	PCWP	CO	SVR	SVO ₂
Hypovolemic shock	Hemorrhage, dehydration, burns	Volume depletion		↓	↓	↓	↑	↓
Cardiogenic shock	MI, HF, valvular dysfunction, arrhythmia	Left heart dysfunction	Cold, clammy	↑	↑	↓	↑	↓
Obstructive shock	PE, tension pneumothorax	Impeded cardiopulmonary blood flow		↑	↓	↓	↑	↓
	Cardiac tamponade			↑	↑	↓	↑	↓
Distributive shock	Sepsis (early), anaphylaxis	Systemic vasodilation	Warm, dry	↓	↓	↑	↓	↑
	CNS injury			↓	↓	↓	↓	normal/↑

↓ = 1° disturbance driving the shock.

Cardiac tamponade



Compression of the heart by fluid (eg, blood, effusions) → ↓ CO. Equilibration of diastolic pressures in all 4 chambers.

Findings: Beck triad (hypotension, distended neck veins, distant heart sounds), ↑ HR, pulsus paradoxus. ECG may show low-voltage QRS and/or electrical alternans (due to “swinging” movement of heart in large effusion). Echocardiogram shows pericardial effusion (arrows in **A**), systolic RA collapse, diastolic RV collapse, and IVC plethora.

Treatment: pericardiocentesis or surgical drainage.

Pulsus paradoxus

↓ in amplitude of systolic BP by > 10 mm Hg during inspiration. ↑ venous return during inspiration → ↑ RV filling → interventricular septum bows toward LV (due to ↓ pericardial compliance) → ↓ LV ejection volume → ↓ systolic BP. Seen in constrictive pericarditis, obstructive pulmonary disease (eg, Croup, OSA, Asthma, COPD), cardiac Tamponade (pea COAT).

Syncope

Transient loss of consciousness caused by a period of ↓ cerebral blood flow. Types:

- Reflex (most common)—vasovagal (prodromal symptoms [eg, warmth, pallor, nausea]; episodes are short with rapid recovery), situational (eg, coughing/sneezing, swallowing, defecation, micturition), carotid sinus hypersensitivity (eg, wearing tight collar).
- Orthostatic—hypovolemia, drugs (eg, antihypertensives), autonomic dysfunction. Orthostatic hypotension is defined as a drop in systolic BP ≥ 20 mm Hg and/or diastolic BP ≥ 10 mm Hg within 3 minutes of standing.
- Cardiac—arrhythmias, structural (eg, aortic stenosis, HCM).

Infective endocarditis

Infection of the endocardial surface of the heart, typically involving ≥ 1 heart valves.

Caused by bacteria $\gg$ fungi. Forms:

- **Acute**—classically *S aureus* (high virulence). Large destructive vegetations **A** on previously normal valves. Rapid onset.
- **Subacute**—classically viridans streptococci (low virulence). Smaller vegetations on congenitally abnormal or diseased valves. Sequela of dental procedures. Gradual onset.

Presents with fever (most common), new murmur, vascular and immunologic phenomena.

Vascular phenomena—septic embolism, petechiae, splinter hemorrhages (linear hemorrhagic lesions on nail bed **B**), Janeway lesions (painless, flat, erythematous lesions on palms or soles).

Immunologic phenomena—immune complex deposition, glomerulonephritis, **Osler** nodes (painful [“**O**uchy”], raised, violaceous lesions on finger or toe pads **C**), **Roth** spots (**R**etinal hemorrhagic lesions with pale centers **D**).

Mitral valve (most common) $>$ aortic valve.

Tricuspid valve involvement is associated with injection **drug** use (don’t “**tri**” **drugs**).

Common associations:

- Prosthetic valves—*S epidermidis*
- GI/GU procedures—*Enterococcus*
- Colon cancer—*S gallolyticus*
- Gram $\ominus$ —**HACEK** organisms (*Haemophilus*, *Aggregatibacter* [formerly *Actinobacillus*], *Cardiobacterium*, *Eikenella*, *Kingella*)
- Culture $\ominus$ —*Coxiella*, *Bartonella*
- Injection drug use—*S aureus*, *Pseudomonas*, *Candida*

Endothelial injury $\rightarrow$ formation of vegetations consisting of platelets, fibrin, and microbes on heart valves $\rightarrow$ valve regurgitation, septic embolism (systemic circulation in left-sided endocarditis, pulmonary in right-sided).

Diagnosis requires multiple blood cultures and echocardiography.



Nonbacterial thrombotic endocarditis

Also called marantic endocarditis. Rare, noninfective. Vegetations typically arise on mitral or aortic valve and consist of sterile, platelet-rich thrombi that dislodge easily. Usually asymptomatic until embolism occurs.

Associated with the hypercoagulable state seen in advanced malignancy (especially pancreatic adenocarcinoma) or SLE (called **Libman-Sacks endocarditis** in this setting).

Rheumatic fever

A nonsuppurative consequence of pharyngeal infection with group A β -hemolytic streptococci. Late sequelae include rheumatic heart disease, which affects heart valves—mitral > aortic >> tricuspid (high-pressure valves affected most). Early valvular regurgitation, late valvular stenosis.

Associated with Aschoff bodies (granuloma with giant cells), Anitschkow cells (enlarged macrophages with ovoid, wavy, rodlike nucleus), $\uparrow$ anti-streptolysin O (ASO) and $\uparrow$ anti-DNase B titers.

Immune mediated (type II hypersensitivity); not a direct effect of bacteria. Antibodies to M protein cross-react with self antigens, often myosin (molecular mimicry).

Treatment/prophylaxis: penicillin.

JONES (major criteria):

Joint (migratory polyarthritides)

♥ (pancarditis)

Nodules in skin (subcutaneous)

Erythema marginatum (evanescent rash with ring margin)

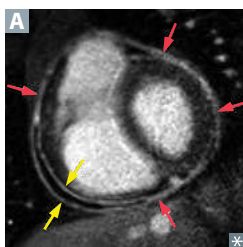
Sydenham chorea (involuntary irregular movements of limbs and face)

Syphilitic heart disease

3° syphilis disrupts the vasa vasorum of the aorta with consequent atrophy of vessel wall and dilation of aorta and valve ring.

May see calcification of aortic root, ascending aortic arch, and thoracic aorta. Leads to “tree bark” appearance of aorta.

Can result in aneurysm of ascending aorta or aortic arch, aortic insufficiency.

Acute pericarditis

Inflammation of the pericardium (red arrows in **A**). Commonly presents with sharp pain, aggravated by inspiration, and relieved by sitting up and leaning forward. Often complicated by pericardial effusion (between yellow arrows in **A**). Presents with friction rub. ECG changes include widespread/diffuse ST-segment elevation and/or PR depression.

Usually idiopathic, but may be due to viral infections (eg, coxsackievirus B), malignancy (metastasis), cardiac surgery, thoracic radiotherapy (early), MI (eg, postcardiac injury syndrome), autoimmune diseases (eg, SLE, drug-induced lupus, rheumatoid arthritis), renal failure (uremia).

Treatment: NSAIDs, colchicine, glucocorticoids, dialysis (uremia).

Constrictive pericarditis

Chronic inflammation of pericardium $\rightarrow$ pericardial fibrosis $\pm$ calcification $\rightarrow$ limited space for expansion $\rightarrow$ $\downarrow$ ventricular filling. Usually idiopathic, but may be due to viral infections, cardiac surgery, thoracic radiotherapy (late). TB is the most common cause in resource-limited countries. $\downarrow$ EDV $\rightarrow$ $\downarrow$ CO $\rightarrow$ $\downarrow$ venous return. Presents with dyspnea, peripheral edema, jugular venous distention, Kussmaul sign, pulsus paradoxus, pericardial knock.

Kussmaul sign

Paradoxical ↑ in JVP on inspiration (normally, inspiration → negative intrathoracic pressure → ↑ venous return → ↓ JVP).

Impaired RV filling → RV cannot accommodate ↑ venous return during inspiration → blood backs up into vena cava → Kussmaul sign. May be seen with constrictive pericarditis, restrictive cardiomyopathy, right HF, massive pulmonary embolism, right atrial or ventricular tumors.

Myocarditis

Inflammation of myocardium. Major cause of SCD in adults < 40 years old.

Presentation highly variable, can include dyspnea, chest pain, fever, arrhythmias (persistent tachycardia out of proportion to fever is characteristic).

Multiple causes:

- Viral (eg, adenovirus, coxsackie B, parvovirus B19, HIV, HHV-6, COVID-19); lymphocytic infiltrate with focal necrosis highly indicative of viral myocarditis
- Parasitic (eg, *Trypanosoma cruzi*, *Toxoplasma gondii*)
- Bacterial (eg, *Borrelia burgdorferi*, *Mycoplasma pneumoniae*, *Corynebacterium diphtheriae*)
- Toxins (eg, carbon monoxide, black widow venom)
- RF
- Drugs (eg, doxorubicin, cocaine)
- Autoimmune (eg, Kawasaki disease, sarcoidosis, SLE, polymyositis/dermatomyositis)

Complications include sudden death, arrhythmias, heart block, dilated cardiomyopathy, HF, mural thrombus with systemic emboli.

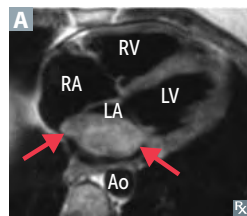
Hereditary hemorrhagic telangiectasia

Also called Osler-Weber-Rendu syndrome. Autosomal dominant disorder of blood vessels. Findings: blanching lesions (eg, tongue telangiectasias **A**) on skin and mucous membranes, recurrent epistaxis, arteriovenous malformations (eg, brain, lung, liver), GI bleeding, hematuria.

Arteriovenous malformation—abnormal, high-flow connection between artery and vein. ↑ risk of high output cardiac failure. Most common cause of intracranial hemorrhage in children.

**Cardiac tumors**

Most common cardiac tumor is a metastasis (eg, melanoma).

Myxomas

Most common 1° cardiac tumor in **adults** (arrows in **A**). 90% occur in the atria (mostly left atrium). Myxomas are usually described as a “ball valve” obstruction in the left atrium (associated with multiple syncopal episodes). IL-6 production by tumor → constitutional symptoms (eg, fever, weight loss). May auscultate early diastolic “tumor plop” sound (mimics mitral stenosis). Histology: gelatinous material, myxoma cells immersed in glycosaminoglycans.

Adults make **6 myxed** drinks.

Rhabdomyomas

Most frequent 1° cardiac tumor in children (associated with tuberous sclerosis). Histology: hamartomatous growths. More common in the ventricles.

Deep venous thrombosis

Blood clot within a deep vein → swelling, redness **A**, warmth, pain. Predisposed by Virchow triad (**SHE**):

- **S**tasis (eg, post-op, long drive/flight)
- **H**ypercoagulability (eg, defect in coagulation cascade proteins, such as factor V Leiden; oral contraceptive use; pregnancy)
- **E**ndothelial damage (exposed collagen triggers clotting cascade)

DVT of proximal deep veins of lower extremity (iliac, femoral, popliteal) → embolic source.

D-dimer test may be used clinically to rule out DVT if disease probability is low or moderate (high sensitivity, low specificity). Imaging test of choice is compression ultrasound with Doppler.

Use unfractionated heparin or low-molecular weight heparins (eg, enoxaparin) for prophylaxis and acute management. Use direct anticoagulants (eg, rivaroxaban, apixaban) for treatment and long-term prevention.

► CARDIOVASCULAR—PHARMACOLOGY**Hypertension treatment**

Primary (essential) hypertension	Thiazide diuretics, ACE inhibitors, angiotensin II receptor blockers (ARBs), Ca ²⁺ channel blockers.	
Hypertension with heart failure	ACE inhibitors/ARBs, β-blockers (compensated HF), aldosterone antagonists (mortality benefit), diuretics (improve symptoms and reduce hospitalization)	β-blockers must be used cautiously in decompensated HF and are contraindicated in cardiogenic shock.
Hypertension with diabetes mellitus	ACE inhibitors/ARBs, Ca ²⁺ channel blockers, thiazide diuretics, β-blockers.	ACE inhibitors/ARBs are protective against diabetic nephropathy. β-blockers can mask hypoglycemia symptoms.
Hypertension with asthma	ARBs, Ca ²⁺ channel blockers, thiazide diuretics, cardioselective β-blockers.	Avoid non-selective β-blockers to prevent β ₂ -receptor–induced bronchoconstriction. Avoid ACE inhibitors to prevent confusion between drug- or asthma-related cough.
Hypertension with pregnancy	H ydralazine, m ethyldopa, l abetalol, n ifedipine.	H ypertensive m oms l ove n ifedipine.
Hypertension with gout	ACE inhibitors, Ca ²⁺ channel blockers, ARBs (especially losartan).	Avoid loop and thiazide diuretics as these can cause gout flares.
Hypertension with osteoporosis	Thiazides preferred as monotherapy; combine with ACE inhibitors or ARBs if needed.	Thiazides preferred due to increased Ca ²⁺ reabsorption.
Hypertension with pheochromocytoma	Phentolamine, phenoxybenzamine (α-blockade first) prior to resection and β-blockade.	Avoid β-blockade first to mitigate unopposed α-mediated hypertensive crisis.